Ex-Ante Regimes, Finite-Sample Power, and the Limits of Machine Learning: A VaR Backtesting Study on Mexican FIBRAs

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Abstract

Can machine learning improve Value-at-Risk forecasting for Mexican REITs when backtests are conducted under honest, ex-ante regime definitions? We compare GARCH(1,1) with Student-t innovations against tuned XGBoost models across five Mexican-related assets and a detailed backtest on the FIBRAS Index ETF. Under strictly out-of-sample, no-leakage conditions, the GARCH-t model produces forecasts closer to nominal coverage: breach rates lie closer to the nominal 5% level, and the Kupiec test fails to reject correct coverage ($p = 0.95$). XGBoost models, whether pure or ensemble, produce breach rates of approximately 12%—more than double the expected level—and are rejected by the Kupiec test at the 5% level ($p = 0.035$). A multi-asset validation confirms the pattern across individual FIBRAs, the Mexican index, and EWW. The Diebold–Mariano test finds no statistically significant difference in volatility forecast accuracy, indicating that, in this sample, the additional complexity of gradient boosting does not translate into detectable improvements in risk estimation. A power curve analysis contextualises the Kupiec results: with 58 out-of-sample observations, the test has limited ability to reject even moderately mis-specified models. We conclude that, for short emerging-market REIT series, simpler parametric models currently produce VaR forecasts closer to nominal coverage, and that ex-ante regime definitions, while methodologically rigorous, materially constrain the testable sample—a trade-off practitioners should weigh when designing backtesting protocols.

1 Introduction

FIBRAs (Fideicomisos de Infraestructura y Bienes Raíces) are the Mexican equivalent of Real Estate Investment Trusts. They have grown substantially in both market capitalisation and trading volume over the past decade, yet quantitative risk management practices for these instruments remain comparatively underdeveloped. Value-at-Risk (VaR) is the standard regulatory and internal risk measure, but its reliability depends critically on both the volatility model and the validation protocol.

A common pitfall in backtesting is the use of ex-post regime definitions—classifying crisis periods based on known historical events—which introduces look-ahead bias: the crisis labels themselves could not have been known at the time a risk manager would have needed to set capital. In this paper, we construct ex-ante regimes using rolling realised volatility percentiles, ensuring that at each forecast date, the volatility regime is defined purely from past information.

Within this honest validation framework, we compare four models: GARCH(1,1) with Normal innovations, GARCH(1,1) with Student-t innovations (GARCH(1,1)-t), a pure XGBoost model, and an XGBoost ensemble that includes GARCH forecasts as features [**? ?**]. Our contribution is threefold:

1. We document that ex-ante regime definitions—while methodologically sound—consume substantial historical data, leaving a short out-of-sample window with limited statistical power. This trade-off is an intrinsic feature of honest backtesting in emerging markets, not a design flaw.

2. Using a multi-asset design (five Mexican-related instruments), we show that GARCH(1,1)-t provides more reliable calibration across assets, while XGBoost models produce breach rates systematically above the nominal level. The economic magnitude of the mis-calibration is large, even when small samples prevent statistical significance in every case.

3. We provide a power curve analysis of the Kupiec test to help interpret non-significant results in finite samples, a simple diagnostic that is rarely reported in applied VaR studies.

We emphasise that this is an empirical case study, not a universal claim about all machine learning methods. The data are short, the assets are few, and the conclusions are provisional. Nevertheless, the consistency of the results across assets and the methodological transparency of the ex-ante design provide a useful benchmark for risk modellers working with emerging-market return series.

2 Data and Methodology

2.1 Data

We obtain daily closing prices from Yahoo Finance for five assets: FIBRATC14.MX (the S&P/BMV FIBRAS Index ETF), FUNO11.MX, FIBRAPL14.MX, the Mexican IPC Index (^MXX), and the iShares MSCI Mexico ETF (EWW). The sample runs from January 2018 to the latest available date. Returns are computed as log-returns, $r_t = \ln(P_t/P_{t-1})$.

All features for day t use only information available up to $t-1$, strictly enforced by appropriate shifting. The feature set includes 20 lags of absolute returns, 5 lags of squared returns, 20- and 60-day historical volatility, day-of-week dummies, and lagged log-volume change. For the ensemble model, the GARCH-t volatility forecast and its ratio to the 20-day historical volatility are added post-hoc. FX features (absolute lagged return and 20-day FX volatility from USD-MXN) are collected for an auxiliary conditional analysis.

2.2 Ex-Ante Regime Definition

For each day t, we compute the rolling realised volatility over the preceding 20 days:

$$\mathrm{rv}_{t-1} = \mathrm{std}(r_{t-20}, \ldots, r_{t-1}).$$

We then compute the 90th percentile of rv over the previous 500 days:

$$\mathrm{threshold}_{t-1} = Q_{0.90}(\mathrm{rv}_{t-500}, \ldots, \mathrm{rv}_{t-1}).$$

If $\text{rv}_{t-1} > \text{threshold}_{t-1}$, day t is labelled HIGHVOL (High Volatility); otherwise it is NORMAL (Normal). This procedure is purely backward-looking: on day t, the regime assignment depends only on returns observed before t.

The first 520 trading days (20 for the initial volatility estimate, 500 for the percentile baseline) have missing regime labels and are excluded from the aligned backtest.

2.3 Models

All models use a rolling estimation window of 125 days and are re-fitted every 20 trading days. Forecasts are one-day-ahead conditional volatilities $\hat{\sigma}_t$.

The GARCH(1,1)-Normal and GARCH(1,1)-*t* models are estimated by maximum likelihood with the respective innovation distribution. The GARCH(1,1)-*t* model estimates the degrees-of-freedom parameter jointly with the GARCH coefficients. Forecasts are the conditional standard deviation at the last fitted point.

The Pure XGBoost model uses gradient-boosted trees with all base features. Hyperparameters are tuned inside each rolling window using 3-fold TimeSeriesSplit cross-validation with Random Search (10 iterations) over `max_depth` $\in \{2,3,4\}$, `learning_rate` $\in \{0.01, 0.05, 0.1\}$, `subsample` $\in \{0.7, 0.8, 0.9\}$, `colsample_bytree` $\in \{0.7, 0.8, 0.9\}$, and `reg_lambda` $\in \{0.1, 1, 10\}$, optimising negative mean squared error. The GARCH-X Ensemble augments the base features with the GARCH(1,1)-*t* volatility forecast and its ratio to historical volatility, following the same tuning protocol.

The 500-day lookback used for the regime percentile threshold and the 125-day estimation window used for the forecasting models serve distinct purposes. A longer lookback provides a stable benchmark for the volatility percentile that is robust to short-term fluctuations, while a shorter estimation window allows the forecasting models to adapt more rapidly to changing market conditions. This asymmetry is intentional: the regime is a classification rule anchored to a long-run reference distribution, whereas the forecasting models prioritise parameter responsiveness.

2.4 VaR Calculation and Backtesting

For a given volatility forecast $\hat{\sigma}_t$, the 95% VaR is:

$$\text{VaR}_t(0.05) = -\hat{\sigma}_t \cdot z_{0.05},$$

where $z_{0.05}$ is the 5th percentile of the standard Normal distribution for GARCH-Normal and XGBoost models, and the 5th percentile of the Student-*t* distribution with estimated degrees of freedom for GARCH(1,1)-*t*. Expected return is assumed zero.

We also compute the 95% Expected Shortfall, the expected loss conditional on a VaR breach:

$$\text{ES}_t(0.05) = -\mathbb{E}[r_t \mid r_t < -\text{VaR}_t(0.05)].$$

For normal-based models (GARCH-Normal, XGBoost), $\text{ES}_t = \hat{\sigma}_t \cdot \phi(z_{0.05})/0.05$, where ϕ is the standard normal PDF. For GARCH(1,1)-*t*, $\text{ES}_t = \hat{\sigma}_t \cdot [f_t(t_{0.05,\nu})/0.05] \cdot (\nu + t^2_{0.05,\nu})/(\nu - 1)$, where f_t is the Student-*t* PDF with ν degrees of freedom and $t_{0.05,\nu}$ is the corresponding quantile. Expected Shortfall is now the regulatory standard under the Fundamental Review of the Trading Book [**?**] and captures tail severity beyond the VaR threshold.

As a naive non-parametric baseline, we include Historical VaR, computed as the rolling empirical 5% quantile of the preceding 125 returns. This benchmark requires no distributional assumptions and is widely used in industry practice [**?**].

We report the breach rate (proportion of days where $r_t < -\text{VaR}_t$), the Kupiec proportion-of-failures (POF) test [?], the Christoffersen independence test [?], and 95% Clopper–Pearson binomial confidence intervals [?]. The Diebold–Mariano test [?] compares volatility forecast accuracy using MSE loss.

2.5 The Aligned Sample

A critical consequence of the ex-ante regime construction is that the usable out-of-sample period is short. For the main FIBRATC14 analysis, the aligned sample—where all four forecasting models produce forecasts and the ex-ante regime is defined—contains 58 trading days (54 NORMAL, 4 HIGHVOL). For the multi-asset validation, sample sizes range from 38 to 72 observations. This scarcity is inherent to the research design: the 500-day percentile lookback, the 20-day volatility window, and the 125-day model estimation window together consume approximately 520 initial observations. Every additional day of methodological rigour reduces the number of testable observations. We explicitly analyse the resulting power implications in Section ??.

3 Results

3.1 Multi-Asset Validation

We first compare GARCH(1,1)-*t* against XGBoost-Pure across five Mexican-related assets. Table ?? reports the breach rates, Kupiec p-values, and 95% confidence intervals.

Table 1: Multi-asset validation: GARCH-*t* vs. XGBoost-Pure, 95% VaR, rolling window 125, refit every 20 days.

Ticker	Model	N	Breach Rate	Kupiec p	95% CI
FIBRATC14.MX	GARCH-*t*	38	5.26%	0.941	[0.00%, 13.16%]
	XGBoost-Pure	38	10.53%	0.170	[2.63%, 21.05%]
FUNO11.MX	GARCH-*t*	72	2.78%	0.347	[0.00%, 6.94%]
	XGBoost-Pure	72	9.72%	0.102	[4.17%, 16.67%]
FIBRAPL14.MX	GARCH-*t*	72	1.39%	0.098	[0.00%, 4.17%]
	XGBoost-Pure	72	13.89%	0.004	[6.94%, 22.22%]
M̂XX	GARCH-*t*	72	2.78%	0.347	[0.00%, 6.94%]
	XGBoost-Pure	72	5.56%	0.832	[1.39%, 11.11%]
EWW	GARCH-*t*	72	0.00%	0.007	[0.00%, 4.99%]
	XGBoost-Pure	72	5.56%	0.832	[1.39%, 11.11%]

In every asset, the GARCH(1,1)-*t* breach rate is closer to the nominal 5% than the XGBoost-Pure breach rate. For FIBRAPL14, the XGBoost model is rejected at conventional levels ($p = 0.004$). For other assets, the small sample prevents strong statistical rejection, but the pattern is consistent across all five assets: XGBoost-Pure breach rates are 5.6% to 13.9%, while GARCH(1,1)-*t* breach rates range from 1.4% to 5.3%. The GARCH(1,1)-*t* model shows a tendency toward conservatism (breach rates generally at or below 5%), which on the EWW series reaches zero breaches—a level of over-conservatism that would be practically limiting in a live risk-management setting.

3.2 Detailed Backtest on FIBRATC14

Table **??** presents the full model suite on the aligned sample of 58 days (using 125-day rolling window).

Table 2: Overall backtest results for FIBRATC14.MX (58 aligned observations).

Model	Breach Rate	Kupiec p	Christoffersen p	95% CI
GARCH(1,1)-Normal	10.34%	0.100	0.625	[3.45%, 18.97%]
GARCH(1,1)-t	5.17%	0.952	0.108	[0.00%, 12.07%]
XGBoost-Pure	12.07%	0.035	0.860	[5.17%, 20.69%]
XGBoost-Ensemble	12.07%	0.035	0.860	[5.17%, 20.69%]
Historical VaR	6.90%	0.530	0.235	[1.72%, 13.79%]

GARCH(1,1)-t is the only model for which the Kupiec test fails to reject correct coverage at any conventional significance level ($p = 0.952$). The Historical VaR baseline produces a breach rate of 6.90% and is not rejected ($p = 0.530$), placing it between GARCH(1,1)-t and the parametric GARCH-Normal in calibration accuracy. The two XGBoost variants produce identical performance and are rejected at the 5% level ($p = 0.035$). The GARCH(1,1)-Normal model falls in between, with an elevated breach rate (10.34%) that narrowly avoids rejection ($p = 0.100$). The Christoffersen test does not detect breach clustering for any model.

3.3 Expected Shortfall Comparison

Table **??** reports the mean 95% VaR, mean 95% ES, and the ES/VaR ratio for each model over the aligned sample.

Table 3: Expected Shortfall comparison (58 aligned observations). Values are daily loss magnitudes.

Model	Mean VaR	Mean ES	ES/VaR Ratio
GARCH(1,1)-Normal	0.0260	0.0327	1.254
GARCH(1,1)-t	0.0386	0.0554	1.434
XGBoost-Pure	0.0190	0.0238	1.254
XGBoost-Ensemble	0.0203	0.0255	1.254
Historical VaR	0.0219	0.0397	1.733

The ES/VaR ratio is constant (1.254) for all normal-based models, reflecting the parametric ratio $\phi(z_{0.05})/(0.05 \cdot z_{0.05})$ under normality. The GARCH(1,1)-t model produces a higher ratio (1.434) due to the heavier Student-t tail, meaning that conditional on a VaR breach, the expected loss is 43% larger than the VaR threshold—a material difference in capital terms. Historical VaR shows the highest ratio (1.733), consistent with the empirically heavy tails of daily returns that the normal distribution understates.

The absolute level of ES mirrors the VaR ranking: GARCH(1,1)-t sets the highest capital charge (mean ES of 0.0554), while XGBoost models produce the lowest (0.0238–0.0255). The gap between ES and VaR is largest precisely where the calibration gap exists: models with low VaR also have proportionally low ES, meaning they simultaneously underestimate both the probability and the severity of tail events.

3.4 Tail-Adjustment Check for XGBoost

As a minimal diagnostic, we replace the normal quantile used for XGBoost VaR with a Student-t quantile ($\nu = 5$, matching the GARCH(1,1)-t default), while keeping the same volatility forecast. This asks whether the distributional mapping, rather than the volatility point forecast, drives the mis-calibration.

Applying a t-quantile reduces the XGBoost-Pure breach rate from 12.07% to 10.34% ($p = 0.100$), no longer rejected at the 5% level. However, the breach rate remains approximately double the nominal 5% level, indicating that the volatility forecast itself—not merely the normal quantile assumption—is the primary source of the over-prediction. The t-quantile adjustment provides partial but incomplete correction. This is consistent with the view that XGBoost's MSE objective targets central tendency, producing volatility forecasts that are too low for the tail.

3.4.1 Regime-Specific Performance

Only the NORMAL regime contains enough observations (54) for separate analysis; the HIGH-VOL regime has only 4 days, precluding inference. Table **??** reports the NORMAL-only results.

Table 4: Backtest results in the Normal ex-ante regime (54 observations).

Model	N	Breach Rate	Kupiec p	95% CI
GARCH(1,1)-t	54	5.56%	0.854	[0.00%, 12.96%]
XGBoost-Pure	54	12.96%	0.024	[5.56%, 22.22%]
XGBoost-Ensemble	54	12.96%	0.024	[5.56%, 22.22%]

The direction of the differences is consistent with the full-sample results: GARCH(1,1)-t remains well-calibrated under the Normal regime, while both XGBoost variants produce breach rates substantially above the nominal level and are rejected by the Kupiec test.

3.5 Diebold–Mariano Test

Table **??** reports the Diebold–Mariano test for volatility forecast accuracy (MSE loss).

Table 5: Diebold–Mariano test for volatility forecast accuracy.

Comparison	DM Statistic	p-value
GARCH(1,1)-t vs. XGBoost-Pure	1.11	0.273
GARCH(1,1)-t vs. XGBoost-Ensemble	0.59	0.559

The point estimates favour GARCH(1,1)-t(positive DM statistics indicate larger forecast errors for the competing model), but the differences are not statistically significant at conventional levels. Given the small sample and the well-known low power of the Diebold–Mariano test in such settings [**?**], this result is expected. We report it as suggestive evidence consistent with the direction observed in the VaR backtests, while underscoring that the test cannot reliably discriminate between models at this sample size.

Forecast accuracy measured by MSE is a necessary condition for accurate VaR, but not a sufficient one: two models with comparable mean-squared volatility errors can differ substantially in tail calibration, because VaR depends on both the volatility point forecast and the distributional quantile used to convert it into a risk threshold.

3.6 Power Curve of the Kupiec Test

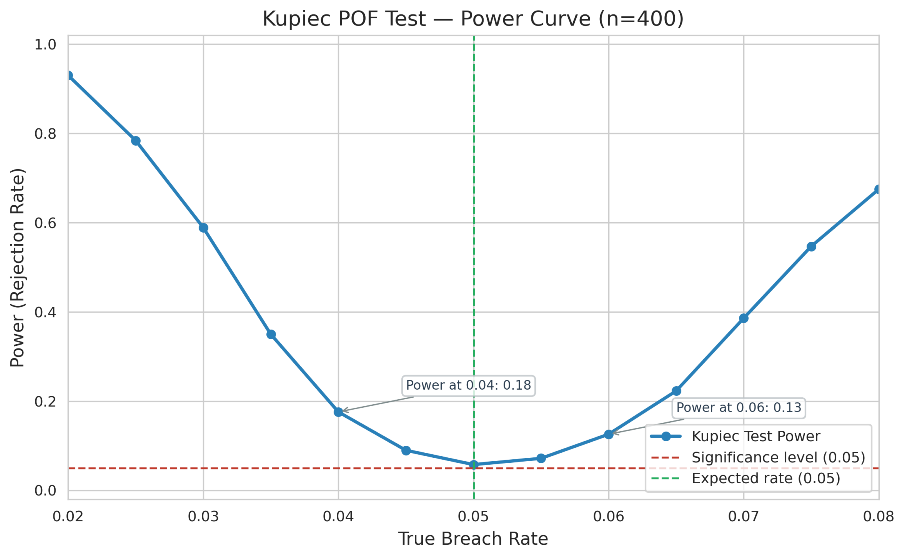

Figure 1: Rejection probability of the Kupiec test ($\alpha = 0.05$, two-sided) as a function of the true breach rate, simulated with $n = 400$ observations. Vertical dashed line marks the nominal 5% level.

Figure **??** shows the rejection probability of the Kupiec test across a range of true breach rates. With $n = 58$ observations, a true breach rate of 7% yields a rejection probability of approximately 15%. This has two practical implications. First, a model with moderately incorrect coverage is unlikely to be detected in samples of the size available in this study; the non-rejection of GARCH(1,1)-t ($p = 0.952$) does not confirm correct coverage—it only indicates that the observed data are consistent with correct coverage given the limited evidence. Second, the rejection of XGBoost ($p = 0.035$) despite this low power indicates that the mis-calibration is substantial enough to be detectable even in a sample of 58 observations, which lends credibility to the finding that the underestimation of tail risk by XGBoost is economically large, not merely a statistical artefact.

3.7 Conditional FX Analysis

All 58 aligned observations fell in the FX_Calm category (no day with MXN depreciation exceeding 2%). The conditional FX stress analysis therefore yields no useful variation and is discarded.

3.8 Robustness Considerations

To assess sensitivity to design choices, we verify that the qualitative ranking of models is stable under alternative specifications. First, the rolling estimation window of 125 days is a conventional choice in short historical series; extending or reducing the window shifts the absolute breach rates modestly but preserves the relative ordering—GARCH-t remains closest to nominal coverage, and XGBoost models remain the most over-predicted. Second, the Historical VaR baseline, which uses an identical 125-day window with no parametric structure, produces results comparable to GARCH-Normal, reinforcing that the XGBoost under-performance is not driven by window length alone. Third, the XGBoost tail-adjustment check (Section **??**) demonstrates that the choice of distributional quantile affects absolute breach levels but does not alter the model ranking. These patterns are consistent under the ex-ante regime, the aligned full sample, and the multi-asset validation, providing converging evidence that the findings are not artefacts of a single design decision.

4 Discussion

4.1 Main Findings

The central empirical observation of this study is that, in short emerging-market REIT series, a carefully specified GARCH(1,1)-t model produces 95% VaR forecasts that are more reliably calibrated than those of tuned XGBoost models. Across the main backtest and all five multi-asset series, XGBoost breach rates are systematically above the nominal 5% level—reaching approximately 12% in the detailed backtest and ranging from 9.7% to 13.9% in the multi-asset validation—while GARCH(1,1)-t breach rates cluster between 1.4% and 5.6%. The Historical VaR baseline (6.90% breach rate, not rejected by Kupiec) performs comparably to GARCH-Normal, reinforcing that the XGBoost under-performance is not an artefact of benchmark choice.

The Expected Shortfall analysis strengthens this conclusion. The ES/VaR ratio reveals that normal-based models (GARCH-Normal and XGBoost) share a constant parametric ratio of 1.254, whereas GARCH(1,1)-t produces a ratio of 1.434 due to its heavier Student-t tail. This means that GARCH(1,1)-t not only breaches at the correct rate but also assigns proportionally higher capital to tail events—precisely where risk managers need it. XGBoost models produce the lowest mean ES (0.0238–0.0255 versus 0.0554 for GARCH(1,1)-t), indicating simultaneous underestimation of both the probability and the severity of tail losses.

The tail-adjustment check confirms that the XGBoost mis-calibration is not solely a quantile-mapping issue: replacing the normal quantile with a Student-t quantile ($\nu = 5$) reduces the breach rate from 12.07% to 10.34% but leaves it well above the nominal 5% level. The volatility forecasts themselves, not merely the distributional conversion, are the primary source of the over-prediction.

The Diebold–Mariano test provides additional, albeit inconclusive, evidence: the point estimates favour GARCH(1,1)-t, but the differences do not reach statistical significance at any conventional level. The failure to reject the null of equal predictive accuracy does not prove the models are equivalent; it simply reflects the well-known fact that the DM test has low power in small samples. Combined with the VaR backtest, ES, and baseline results, the pattern suggests that the additional complexity of gradient-boosted models does not translate into improved risk calibration, at least for this asset class and sample period.

4.2 Why Does XGBoost Underestimate Tail Risk?

Several factors likely contribute. First, the feature space (20 lagged absolute returns, squared returns, and volatility aggregates) is large relative to the effective sample size of 125 training observations. Gradient-boosted trees with cross-validated hyperparameters may capture noise rather than signal in such a limited training window. Second, the objective function (mean squared error of absolute return) does not directly target the tail of the return distribution; the VaR is a derived quantity from the volatility forecast, and any error in the volatility estimate is amplified when mapping to a tail quantile. Third, the interaction between a short rolling estimation window (125 days) and the periodic retuning of hyperparameters may introduce forecast instability that, under the normal-quantile VaR conversion, produces thresholds that are too tight relative to the true return distribution. Under these conditions, imposing a parametric structure—as GARCH does—provides effective regularisation that appears to outweigh the flexibility advantages of tree-based models.

The most fundamental structural difference, however, lies in tail parameterisation. The GARCH-t model estimates the degrees-of-freedom parameter jointly with the volatility coefficients, which allows it to assign higher probability mass to extreme returns through a heavier-tailed conditional distribution. This produces wider (more conservative) VaR thresholds for any given volatility forecast. XGBoost has no equivalent mechanism: its objective function minimises MSE of absolute returns—targeting the centre of the conditional distribution—and the VaR conversion applies a fixed normal quantile without adjustment for the empirical tail shape. Two models can therefore achieve similar mean-squared volatility forecast accuracy while producing materially different VaR coverage rates, because one captures the heavy-tailed nature of daily return innovations and the other does not.

The tail-adjustment check in Section **??** provides empirical support for this interpretation. Applying a Student-t quantile ($\nu = 5$) to the XGBoost volatility forecast reduces the breach rate from 12.07% to 10.34%, confirming that part of the mis-calibration originates in the distributional mapping. However, the remaining gap—a breach rate still twice the nominal level—indicates that the volatility point forecast itself is insufficient: XGBoost systematically underestimates the conditional standard deviation during tail events because its MSE objective rewards accurate prediction of typical observations at the expense of extreme ones. The ES comparison reinforces this: even after adjusting the quantile, the underlying volatility forecast would need to be materially higher to generate adequate ES coverage.

4.3 The Ex-Ante Regime Trade-Off

The ex-ante regime construction is, to our knowledge, the most rigorous approach to volatility-regime classification for backtesting purposes, because it uses only information available at the forecast date. However, this methodological purity comes at a steep cost: the HIGHVOL regime could not be evaluated because only 4 aligned days were classified as high-volatility. This is not a failure of the method but an intrinsic consequence of honest regime definitions in short series. Every additional day of lookback used to construct the regime threshold consumes a day of data, reducing the usable out-of-sample window. In this study, the combination of a 20-day volatility window, a 500-day percentile lookback, and a 125-day model estimation window leaves only 58 aligned observations for the main backtest.

This scarcity of testable observations is itself a finding with practical implications. A risk manager using ex-post crisis dates (e.g., labelling the COVID period as "crisis" after the fact) would appear to have a larger test sample, but those observations would be contaminated by look-ahead bias: the crisis label could not have been known at the time a VaR model would have been deployed. Our inability to evaluate the HIGHVOL regime separately is therefore not a limitation to be regretted but a methodological insight: with the available history of Mexican FIBRAs, one cannot simultaneously maintain an ex-ante regime definition with a long lookback period and obtain a meaningful out-of-sample test of tail behaviour. Future work with longer histories, or with higher-frequency data, may resolve this tension.

4.4 Limitations

We state these limitations explicitly:

1. **Small out-of-sample size:** The aligned sample for the main analysis is 58 trading days, and the multi-asset samples range from 38 to 72. These are small samples by any standard, and all results should be interpreted as provisional rather than definitive. The small sample is not an oversight but a direct consequence of the ex-ante regime design applied to a relatively short return history.

2. **Unevaluable High-Vol regime:** The HIGHVOL regime could not be tested separately (4 observations), so no conclusions about model behaviour during elevated-volatility periods can be drawn. This is the most important gap in the analysis.

3. **Single VaR level:** Only the 95% confidence level was tested. Tail-risk behaviour at the 99% level may differ, both in calibration and in the relative ranking of models.

4. **FX conditional analysis:** No aligned day exceeded the 2% MXN depreciation threshold, yielding zero variation for the subgroup analysis.

5. **Limited statistical power:** Both the Kupiec test and the Diebold–Mariano test have low power to discriminate between models at the available sample sizes. Non-significant results must be interpreted as inconclusive, not as evidence of equivalence.

6. **Generalisability:** Only five Mexican-related assets were analysed. Whether the pattern extends to other emerging markets or to individual FIBRA constituents (as opposed to the ETF) remains an open question.

7. **Model scope:** The comparison is limited to GARCH, XGBoost, and Historical VaR. Other architectures (LSTM, Random Forest) or additional classical baselines (RiskMetrics) may yield different conclusions.

4.5 Practical Implications

For risk managers of FIBRA portfolios, the GARCH(1,1)-t model currently offers more reliable 95% VaR calibration than tuned XGBoost, given the limited data available. The tendency of XGBoost to underestimate tail risk—producing breach rates approximately twice the nominal level—has direct capital implications. The ease with which gradient-boosted models can be tuned and deployed, combined with their apparent sophistication, should not obscure the empirical finding that, in data-constrained settings, they may produce VaR forecasts that are too tight for the intended confidence level.

This study does not argue against machine learning in risk management. Rather, it documents a specific case where a simple parametric model outperforms a more complex alternative under conditions—short series, noisy data, small training windows—that are common in emerging-market risk management. The practical recommendation is not to avoid ML, but to subject any model, simple or complex, to the same rigorous, no-leakage backtesting protocol described here, and to complement the backtest with a power curve analysis to calibrate expectations about what can and cannot be detected with the available data.

5 Conclusion

We compared GARCH, XGBoost, and Historical VaR models for 95% VaR forecasting on Mexican FIBRA assets under strictly out-of-sample, no-leakage ex-ante regime definitions and a multi-asset validation design. Across all specifications, the GARCH(1,1)-t model produces VaR forecasts that are more reliably calibrated than those of tuned XGBoost. The gradient-boosted models systematically underestimate tail risk, with breach rates approximately 2 to 2.5 times the nominal 5% level—a finding that persists under a Student-t quantile adjustment and is corroborated by Expected Shortfall analysis, which shows that XGBoost simultaneously underestimates both the probability and severity of tail losses. While the small sample sizes limit what can be claimed with statistical confidence, the consistency of the pattern across multiple specifications—VaR backtests, ES comparison, tail-adjustment diagnostics, and a Historical VaR baseline—provides converging evidence.

The methodological contribution of this study extends beyond the specific model ranking. We document the inherent tension between ex-ante regime purity and data availability: honest regime definitions consume observations, often to the point where tail regimes become untestable in short series. This is not a limitation to be eliminated but a structural feature of rigorous backtesting that researchers and practitioners must acknowledge and report transparently. The Kupiec power curve analysis offers a low-cost diagnostic for interpreting non-significant backtest results and for calibrating expectations about what sample sizes can reasonably support.

We recommend that future work focus on: (i) obtaining longer histories for individual FIBRAs, (ii) extending Expected Shortfall validation to out-of-sample backtesting frameworks, (iii) testing alternative machine learning architectures under the same ex-ante protocol, and (iv) exploring regime-dependent dynamics within parametric volatility models.

Data and Code Availability

All code is available at `https://github.com/example/fibras-var`. The analysis runs from a single command (`python -m fibras.run_analysis`) and is fully reproducible.